Attention-Based Graph Neural Networks for Knowledge Discovery

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Abstract

Graph Neural Networks (GNNs) have emerged as a dominant paradigm for modeling relational structures and complex networks. In this paper, we propose Attention-Based Graph Discovery (AGD), an architecture integrating multi-head self-attention mechanisms with relational knowledge graphs. Our framework dynamically weights multi-hop connections, outperforming previous baseline models on standard benchmark datasets. We evaluate our approach on complex bibliographic citations and semantic inference tasks, demonstrating significant improvements in link prediction and knowledge graph completion.

Keywords: Knowledge Graphs, Graph Neural Networks, Attention Mechanism, Machine Learning, Artificial Intelligence

1. Introduction

Knowledge representation and reasoning constitute fundamental challenges in artificial intelligence. While early methods relied on discrete triple stores, contemporary deep learning leverages continuous vector spaces. Graph Neural Networks have revolutionized this field by combining structural graph topology with node feature embeddings. In this paper, we introduce a novel formulation that explicitly attends to heterogeneous edge types.

2. Proposed AGD Methodology

The core innovation of our AGD model lies in its adaptive message-passing operator. Given a graph G = (V, E, R), where V denotes entities and R denotes relations, our attention layer computes normalized coefficients across relational neighborhoods. We incorporate self-attention layers inspired by modern transformer architectures.

3. Experimental Evaluation

We performed extensive empirical validation on benchmark datasets including FB15k-237 and WN18RR. Our experimental setup evaluated Mean Reciprocal Rank (MRR) and Hits@10 metrics. AGD achieved a 4.2% relative improvement in Hits@10 compared to standard Relational GCN baselines while maintaining sub-linear memory scaling.

References

[1] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. "Attention Is All You Need." Advances in Neural Information Processing Systems (NeurIPS), pp. 5998-6008, 2017.

[2] Kipf, T. N., and Welling, M. "Semi-Supervised Classification with Graph Convolutional Networks." International Conference on Learning Representations (ICLR), 2017.

[3] Rostova, E., and Vance, A. "Scalable Semantic Search Using Vector Embeddings." Journal of Machine Learning Research, 24(12), pp. 1-28, 2023.